

Lecture 6: Continuous Random Variables, PDFs, CDFs

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- 1 From PMFs to PDFs::: **{.callout-note icon=false}## Definition: PDF and CDF**A continuous r.v. X has PDF $f_X \geq 0$ with $\int_{-\infty}^{\infty} f_X = 1$, and

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

The CDF is $F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$, with $f_X = F'_X$ where differentiable.... **{.callout-warning icon=false}## Fact** $E[X] = \int x f_X(x) dx$, $\text{Var}(X) = \int (x - E[X])^2 f_X(x) dx$...**Uniform and Exponential: theory vs. simulation** 1

—title: “Lecture 6: Continuous Random Variables, PDFs, CDFs”author: “Jamie Haddock”format: revealjs: output-file: Lecture6_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy import statsfig, axes = plt.subplots(1, 2, figsize=(11, 4))xs = np.linspace(-0.5, 5, 400)for
```

Tuesday we meet the most important continuous distribution of all: the Normal.## Reading & HomeworkReading: §3.1-3.2 (Bertsekas & Tsitsiklis, Introduction to Probability).